PROJECT PROPOSAL

AI-Based Plant Disease Detection App

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Academic Project Proposal

1. Introduction

Agriculture is an important part of the economy and daily life. Plant diseases can reduce crop quality and production, especially when farmers cannot identify a disease at an early stage. This project proposes an AI-Based Plant Disease Detection App that helps users identify possible plant diseases by uploading or capturing a picture of a plant leaf. The application will analyze the image using an artificial intelligence model and show a predicted disease category along with basic guidance.

2. Problem Statement

Plant diseases are often identified through visual inspection, which can be difficult for people who do not have specialist knowledge. Incorrect or delayed identification may lead to crop damage and unnecessary use of treatments. There is a need for an easy-to-use digital tool that can provide an initial, quick indication of a plant's health based on a leaf image.

3. Proposed Solution

The proposed app will allow a user to take a photo or select an existing image of a plant leaf. An AI image classification model will process the image and predict whether it appears healthy or shows signs of one of the diseases supported by the model. The app will display the result and general care recommendations. The output will be presented as an initial screening result, not a guaranteed professional diagnosis.

4. Objectives

- Develop a simple and user-friendly mobile or web-based application.
- Use an AI / machine-learning model to classify plant leaf images.
- Display the predicted disease or healthy-plant result clearly.
- Provide basic prevention and care information relevant to the prediction.
- Help users recognize possible plant diseases earlier and make informed next steps.

5. Main Features

- **Image Upload / Camera:** Users can upload a leaf photo or capture one using a device camera.
- **AI-Based Prediction:** The trained model analyzes the image and predicts a supported class.
- **Result Screen:** The app shows the predicted disease or healthy status, with a confidence score if available.
- **Care Guidance:** General prevention and plant-care suggestions are shown for supported results.
- **Simple Interface:** Clear screens and instructions make the app easy to use.

6. Tools and Technologies

The expected tools include Python for model development, TensorFlow or Keras for image classification, OpenCV or Pillow for image preprocessing, and Flask for a web application interface. HTML, CSS, and JavaScript may be used for the front end. A suitable public plant-leaf image dataset can be used for training and testing. Final tools may be adjusted according to project requirements and available resources.

7. Methodology

First, a suitable labeled dataset of healthy and diseased plant-leaf images will be collected. Next, images will be cleaned, resized, and divided into training, validation, and testing sets. An image-classification model will then be trained or fine-tuned and evaluated using test images. After model evaluation, it will be integrated into the app. Finally, the app will be tested for usability, prediction behavior, and common errors.

8. Expected Outcomes

The expected outcome is a working prototype that accepts plant-leaf images and returns a predicted category for the disease classes included in the training data. It should offer an accessible interface and basic supporting guidance. Prediction quality will depend on image quality, the dataset, and the plant diseases covered by the model.

9. Scope and Limitations

The first version will focus on selected plant types and disease categories supported by the available dataset. It may not reliably identify every disease, pest, nutrient deficiency, or condition in real-world settings. Users should consult an agriculture expert for serious crop problems and before making important treatment decisions.

10. Conclusion

The AI-Based Plant Disease Detection App aims to combine image processing and artificial intelligence to make early plant-disease screening easier. The project is practical for a final-year academic project and can be expanded in the future with more plant types, additional languages, and a larger disease database.